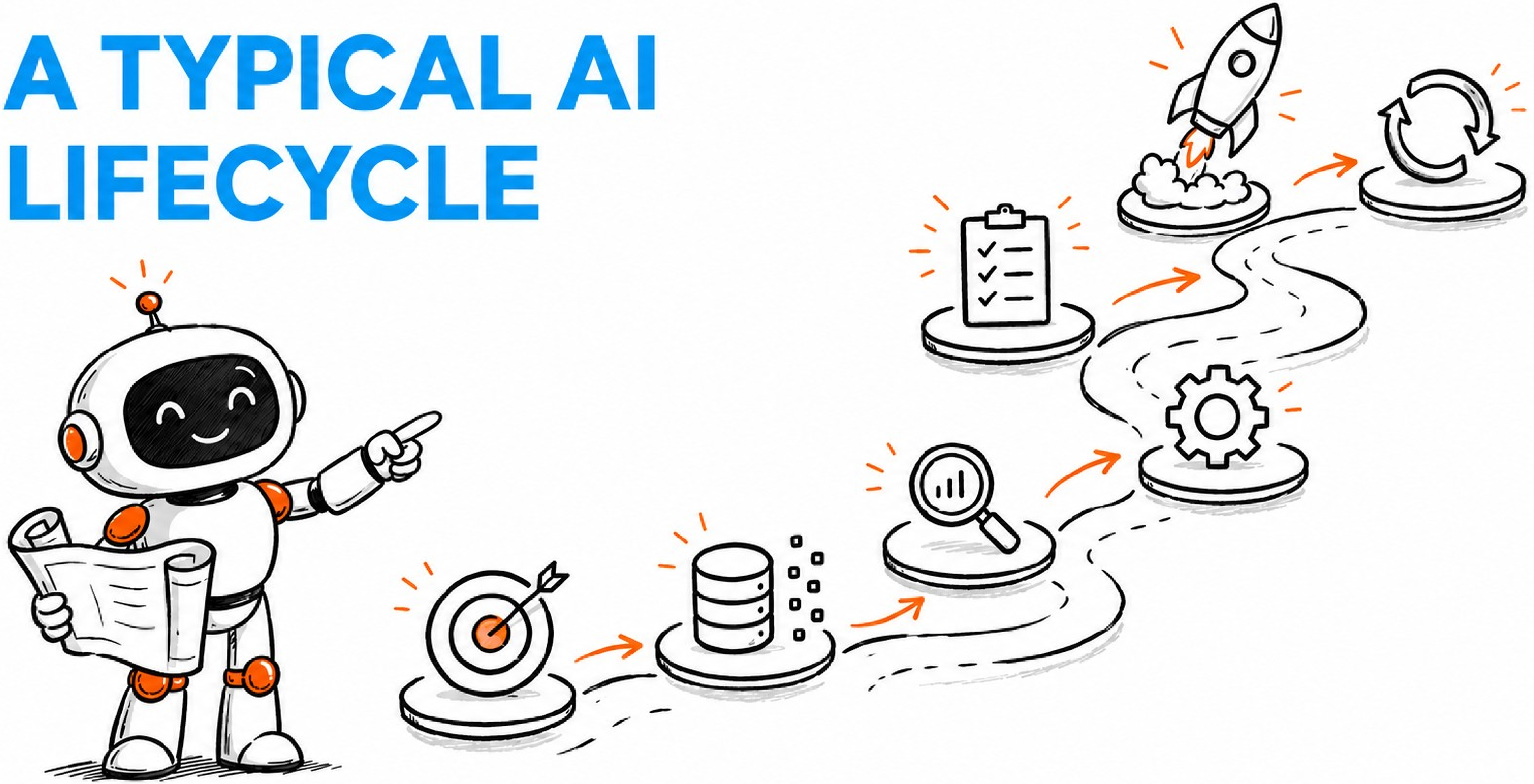
A TYPICAL AI LIFECYCLE



AI Project lifecycle

Step 1. Problem Scoping
Define the business problem & objectives. Understand context & expected outcomes.

Step 2. Data Acquisition
Gather relevant data from internal and/or external sources. This includes structured & unstructured data collection through APIs, databases, sensors, etc.

Step 3. Data Exploration
Analyze the data to understand patterns, identify outliers, handle missing values, & perform feature engineering.

Step 4. Modeling
Develop & train AI models using selected algorithms. Fine-tune the model to improve accuracy & performance.

Step 5. Model Evaluation
Assess model performance using metrics such as accuracy, precision, recall, F1-score, etc. This helps ensure the model is reliable and generalizable.

Step 6. Deployment
Integrate the trained model into a production environment where it can be used by end users to generate predictions.

Step 7. Maintenance & Monitoring
Continuously monitor the model's performance and retrain it as needed to ensure it adapts to new data and remains effective over time.

AI Project Lifecycle

1. Problem Scoping

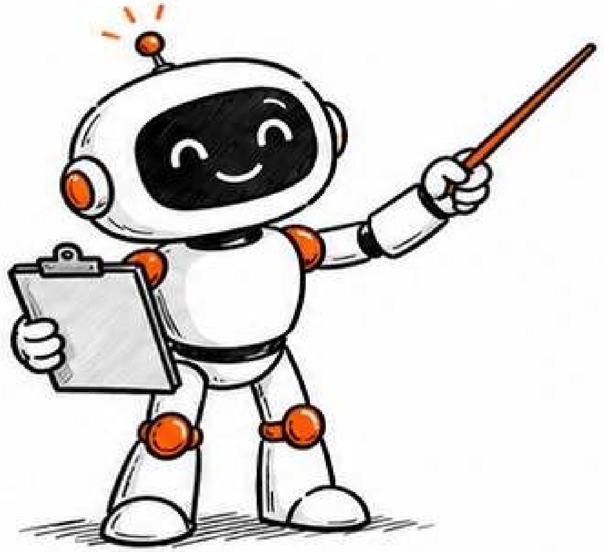
Define the business problem, objectives, context and expected outcomes.

2. Data Acquisition

Gather structured and unstructured data from APIs, databases, sensors and internal or external sources.

3. Data Exploration

Understand patterns and outliers; handle missing values and perform feature engineering.



4. Modeling

Develop and train AI models using selected algorithms; fine-tune for performance.

5. Model Evaluation

Assess accuracy, precision, recall and F1-score to check reliability and generalisability.

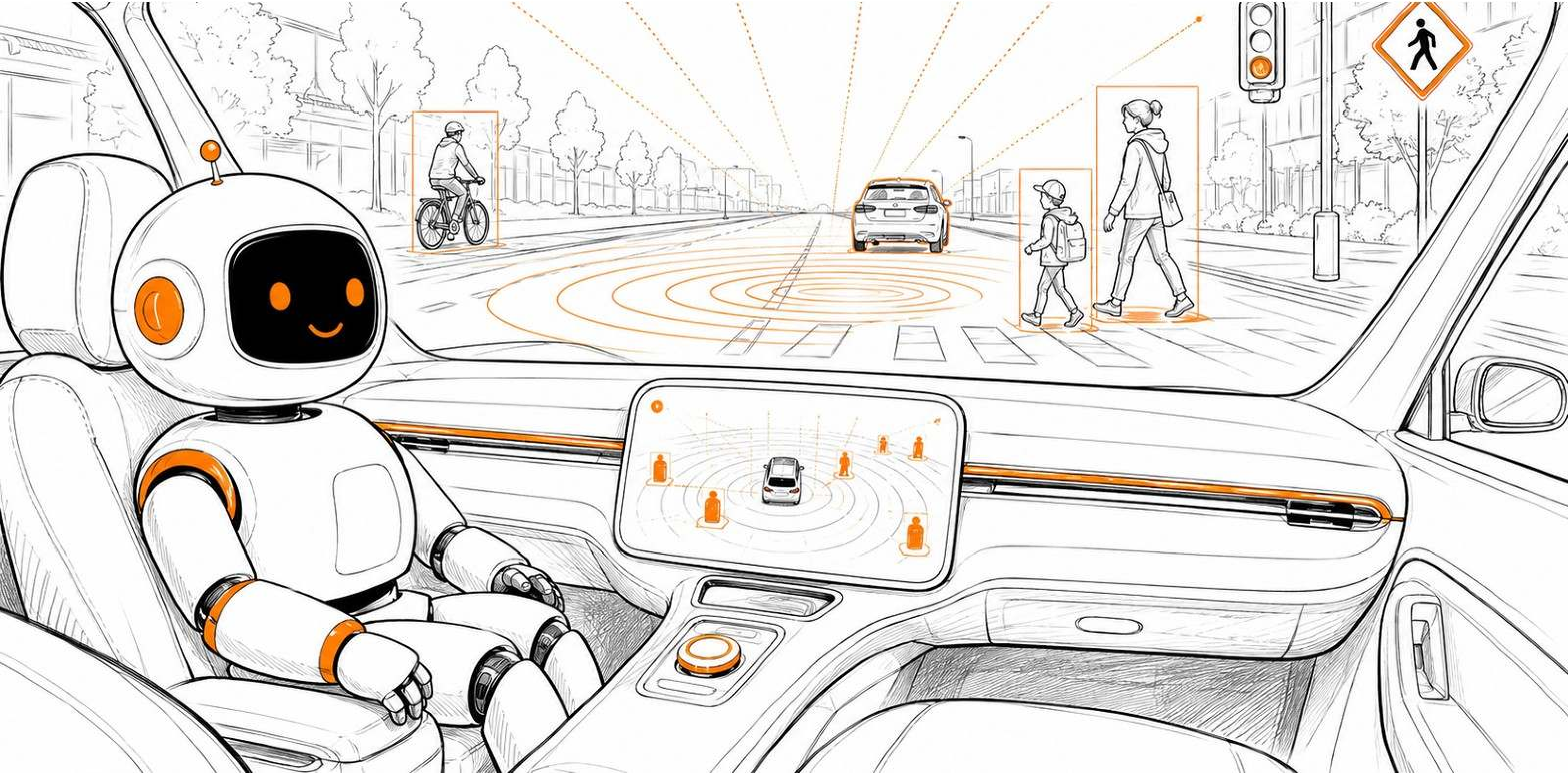
6. Deployment

Integrate the trained model into production so end users can generate predictions.

7. Maintenance & Monitoring

Continuously monitor performance, retrain when needed, and adapt to new data over time.

Practical Example: Self-driving Cars



Practical Example

Step	Question	Answer
1. Problem Scoping	What is the problem you are trying to solve, Where are we deploying them?	Enable a car to perceive its environment and make safe driving decisions. Reduces accidents, improves mobility, and supports autonomous transportation.
2. Data Acquisition	What data would you need to collect to build this AI system?	Camera images, LiDAR/radar data, GPS/IMU readings, and labeled data (e.g., pedestrians, lanes, stop signs). Use datasets like KITTI or Waymo Open Dataset.
3. Data Exploration	What steps would you take to understand and prepare the data?	Visualize sensor inputs, clean noisy data, check for edge cases (night, fog, rain), handle class imbalance, and normalize/resize images.
4. Modeling	What kind of model could you use?	Use Convolutional Neural Networks (CNNs) for vision tasks (e.g., pedestrian detection), sensor fusion models for perception, or reinforcement learning for decision-making (e.g., driving policy).
5. Model Evaluation	What metrics would you use, and why?	Use accuracy, precision, recall, mAP for object detection, IoU for bounding boxes, and latency for real-time performance. Prioritize recall for safety-critical objects.
6. Deployment	Where and how would this AI model be deployed?	Deploy on an edge device (NVIDIA GPU) inside the car. Integrate with the car’s control system to make real-time driving decisions offline.
7. Maintenance & Monitoring	How will you maintain and monitor the model over time?	Continuously collect new driving data, retrain with edge cases, simulate updates before rollout, check for model drift, and monitor fairness across environments.

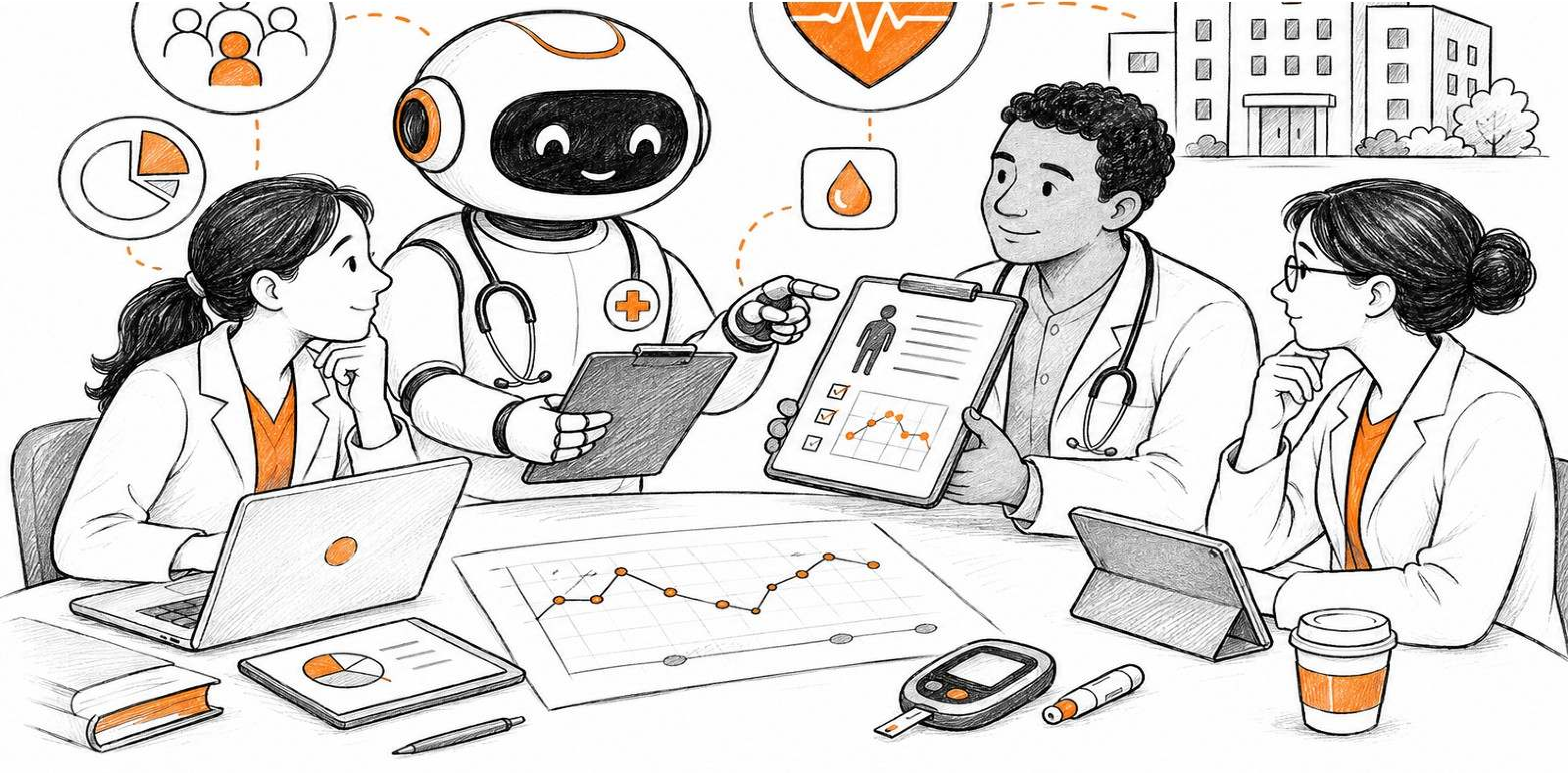
PRACTICE OPPORTUNITY

Practice Opportunity

You're responsible for designing an AI system to predict whether a patient is at risk of developing diabetes based on their health data. Apply the 7 steps of the AI lifecycle to plan this project.
For each step, think critically about what actions you would take, what information you need, and how each decision supports the goal of building a reliable, real-world AI solution.

1. Problem Scoping: What is the problem you are trying to solve, and why is it important?
2. Data Acquisition: What data would you need to collect to build this AI system?
3. Data Exploration: What steps would you take to understand and prepare the data?
4. Modeling: What kind of model could you use to predict diabetes risk?
5. Model Evaluation: What metrics would you use to evaluate your model, and why?
6. Deployment: Where and how would this AI model be deployed?
7. Maintenance & Monitoring: How will you maintain and monitor the model over time?

Practice Opportunity: Diabetes Risk

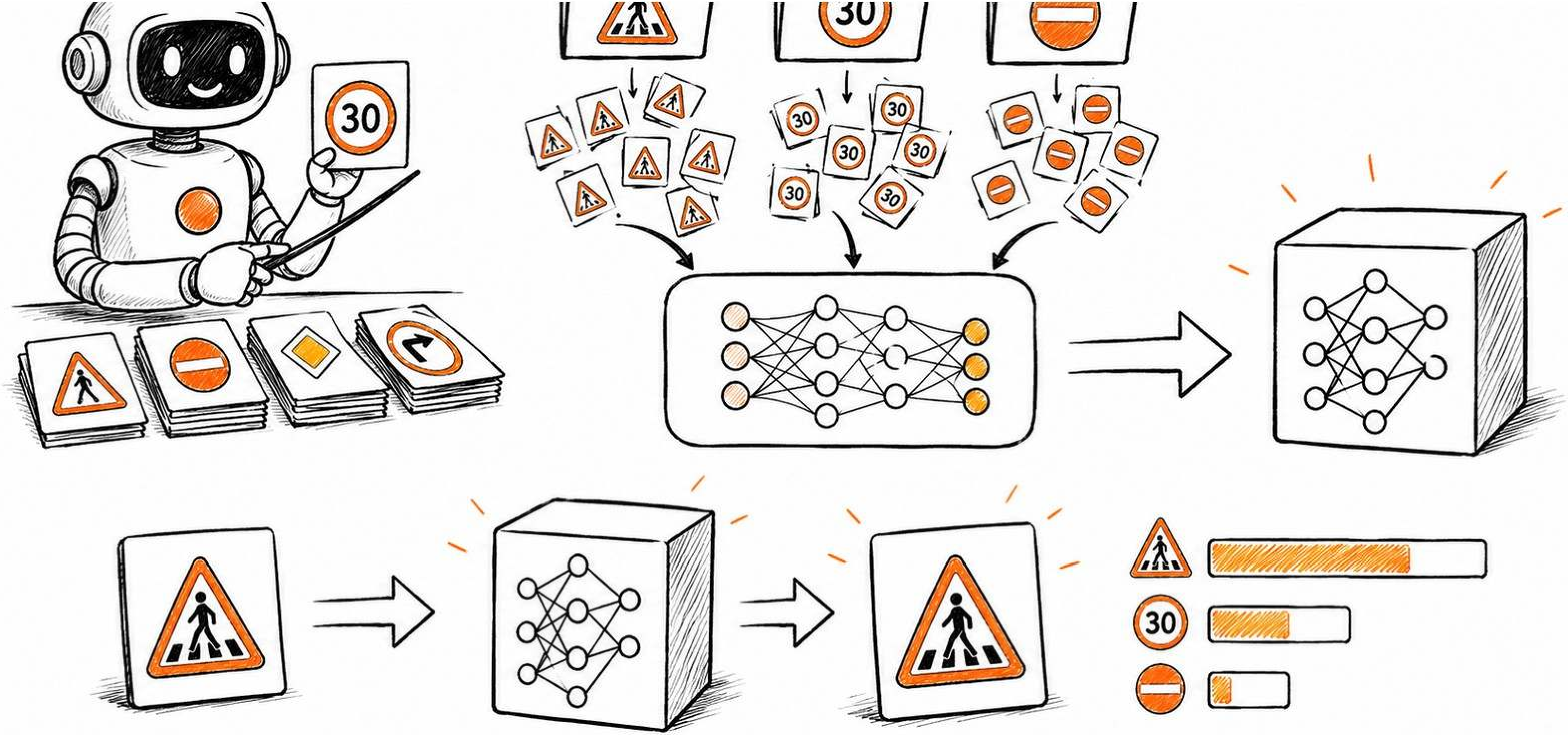


Practice Opportunity Solution

Step	Question	Solution
1. Problem Scoping	What is the problem you are trying to solve, and why is it important?	The goal is to predict if a patient is at risk of diabetes. Early detection helps doctors intervene sooner and improve patient outcomes.
2. Data Acquisition	What data would you need to collect to build this AI system?	Patient data such as age, gender, BMI, blood pressure, glucose levels, insulin levels, physical activity, and family history of diabetes.
3. Data Exploration	What steps would you take to understand and prepare the data?	Check for missing values, outliers, and imbalanced classes. Visualize relationships (e.g., high glucose vs. diabetes) and clean the data.
4. Modeling	What kind of model could you use to predict diabetes risk?	Use a classification model such as logistic regression, decision tree, or random forest.
5. Model Evaluation	What metrics would you use to evaluate your model, and why?	Accuracy, precision, recall, F1 score. Recall is especially important to avoid missing high-risk patients (false negatives).
6. Deployment	Where and how would this AI model be deployed?	Deploy in a hospital’s Electronic Health Record system to provide real-time diabetes risk predictions to doctors.
7. Maintenance & Monitoring	How will you maintain and monitor the model over time?	Regularly monitor performance, retrain with new data, check for model drift, and ensure fairness across patient demographics.

BUILD A SIMPLE AI MODEL & KEY DEFINITIONS

Build and Test a Simple AI Model



Key AI Components

Data + **Algorithms (Models)** + **Computing Power**

Data

The foundation of AI, as it requires large amounts of data for training.

Algorithms (Models)

Mathematical models that process data to recognize patterns & make decisions.

Computing Power

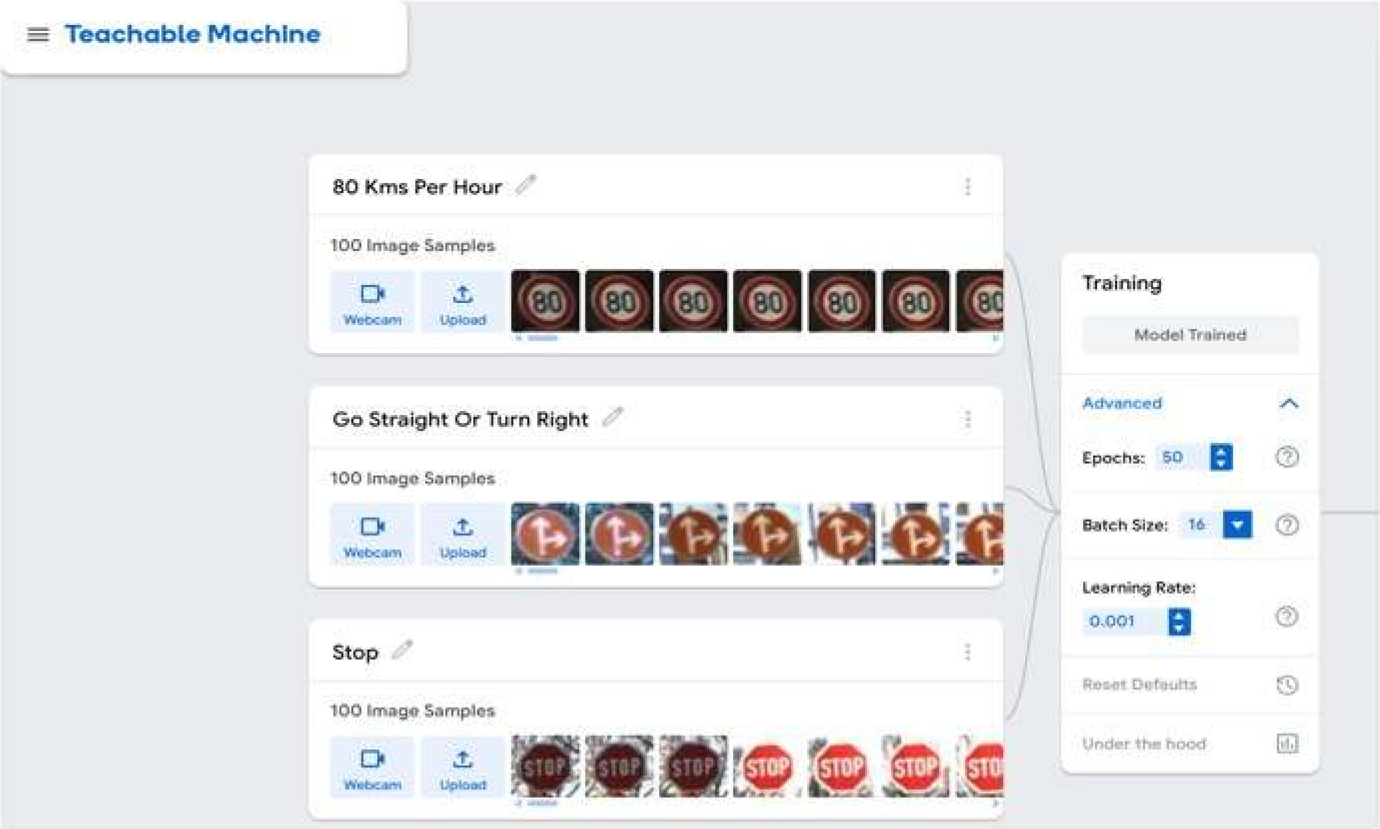
Computational resources (e.g.: GPUs) to handle large datasets & run complex algorithms.

Training a Simple AI Model

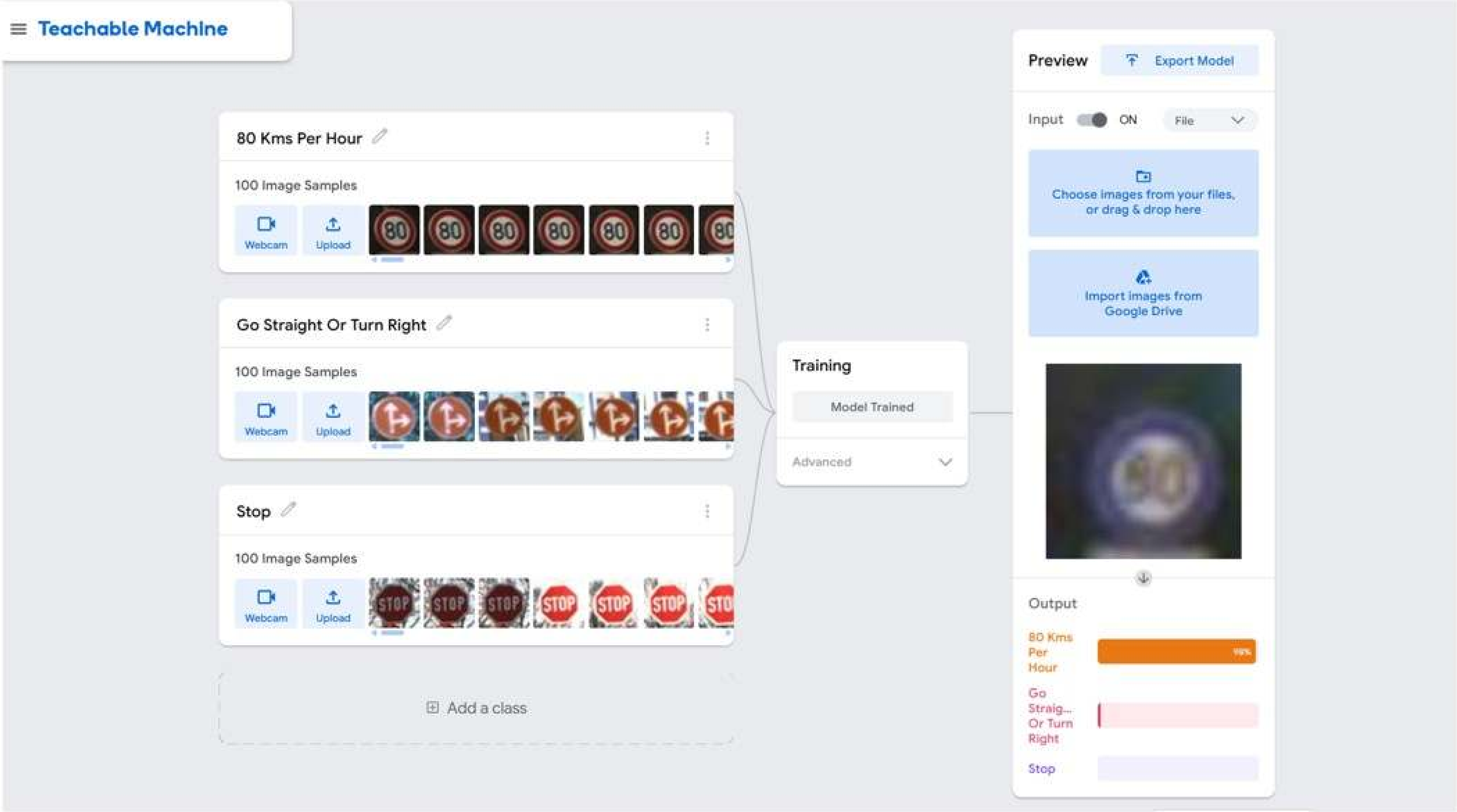
In this example, let's train a simple AI model to classify traffic signs using Google's Teachable Machine; a no-code tool that makes it easy to build AI models. We have 3 classes with 100 images from each class.

- "80 Kms Per Hour" – 100 images of the 80 km/h sign.
- "Go Straight Or Turn Right" – 100 images of that sign.
- "Stop" – 100 images of stop signs.

Google Teachable Machines: https://teachablemachine.withgoogle.com/



Model Testing



Key Definitions

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Term	Meaning
Model	The AI system that learns from the image samples
Training	The process of teaching the model to recognize patterns
Epochs	Number of full passes over the training dataset
Batch Size	Number of data samples processed at a time
Learning Rate	Speed at which the model adjusts during training

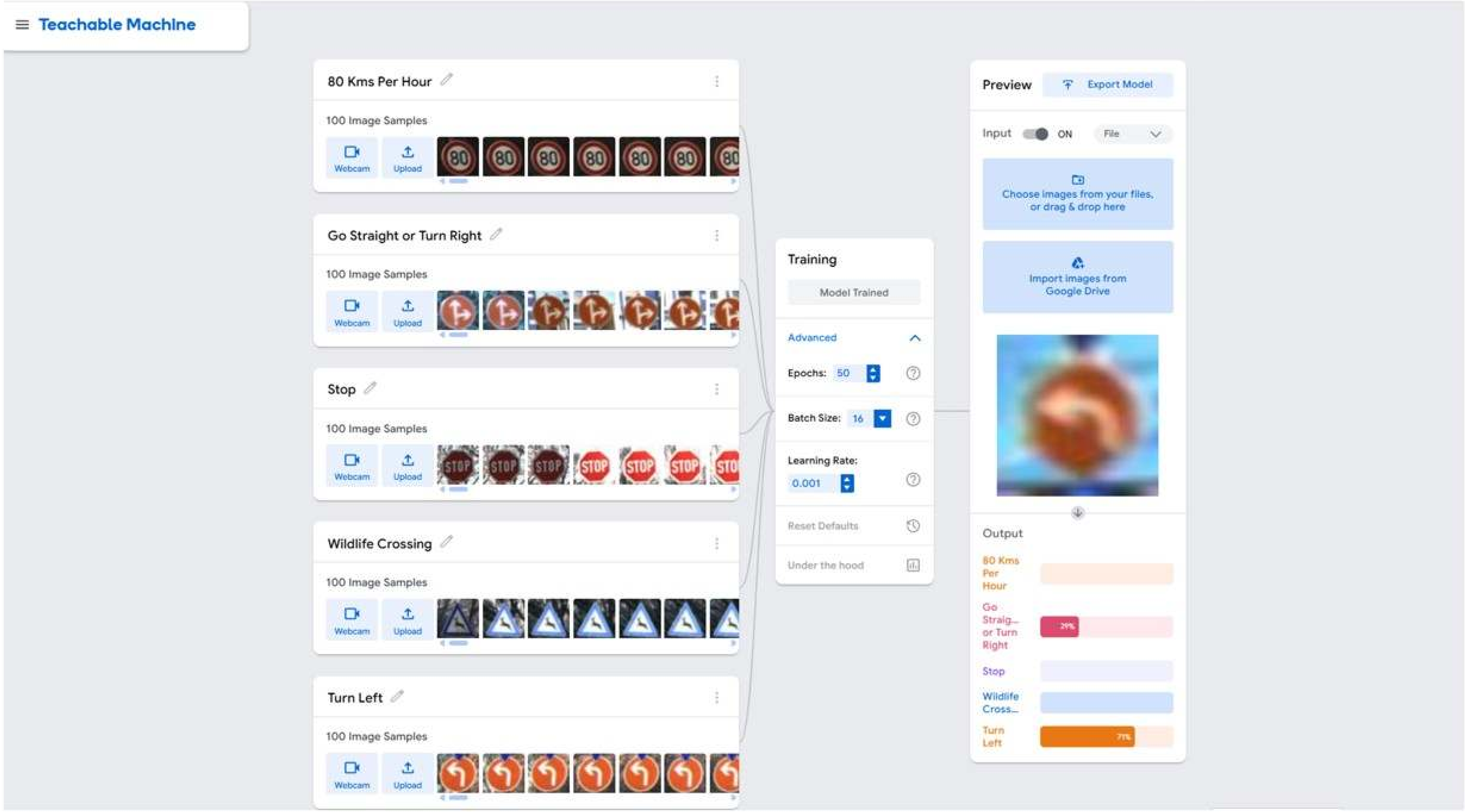
PRACTICE OPPORTUNITY

Practice Opportunity

Using Google Teachable Machines, train and evaluate an image classification model that can recognize two additional traffic signs: (1) Turn Left, and (2) Wildlife Crossing. Perform the following tasks:

- **Data Collection:** Access the training and testing datasets for both classes from the “Practice Opportunity Data” folder.
- **Model Development:** Use Google Teachable Machine to create an image classification model. Train the model using the platform’s built-in tools.
- **Model Evaluation:** Test your model using 3 sample testing datasets for each traffic sign class. Record the model’s predictions and analyze its performance.

Practice Opportunity Solution



Key Challenges in AI Projects

Key Challenges in AI lifecycle steps

1. Problem Scoping:
Unclear objectives or business goals, Lack of stakeholder alignment, Difficulty in translating business needs into AI problems

2. Data Acquisition:
Insufficient or low-quality data, Data access restrictions, Data privacy and compliance issues (e.g., GDPR)

3. Data Exploration:
Missing, noisy, or imbalanced data, Data bias affecting model fairness, High effort required for data cleaning & preprocessing

4. Modeling:
Choosing the right algorithm for the task, Overfitting or underfitting the model, High computational cost & time during training

5. Model Evaluation:
Inadequate evaluation metrics, Lack of domain-specific benchmarks, Risk of data leakage skewing performance

6. Deployment:
Integration complexity with existing systems, Latency & scalability concerns, ensuring real-time inference capability

7. Maintenance & Monitoring Model drift as data patterns change, Lack of continuous monitoring pipelines, Difficulty in automated retraining & updating

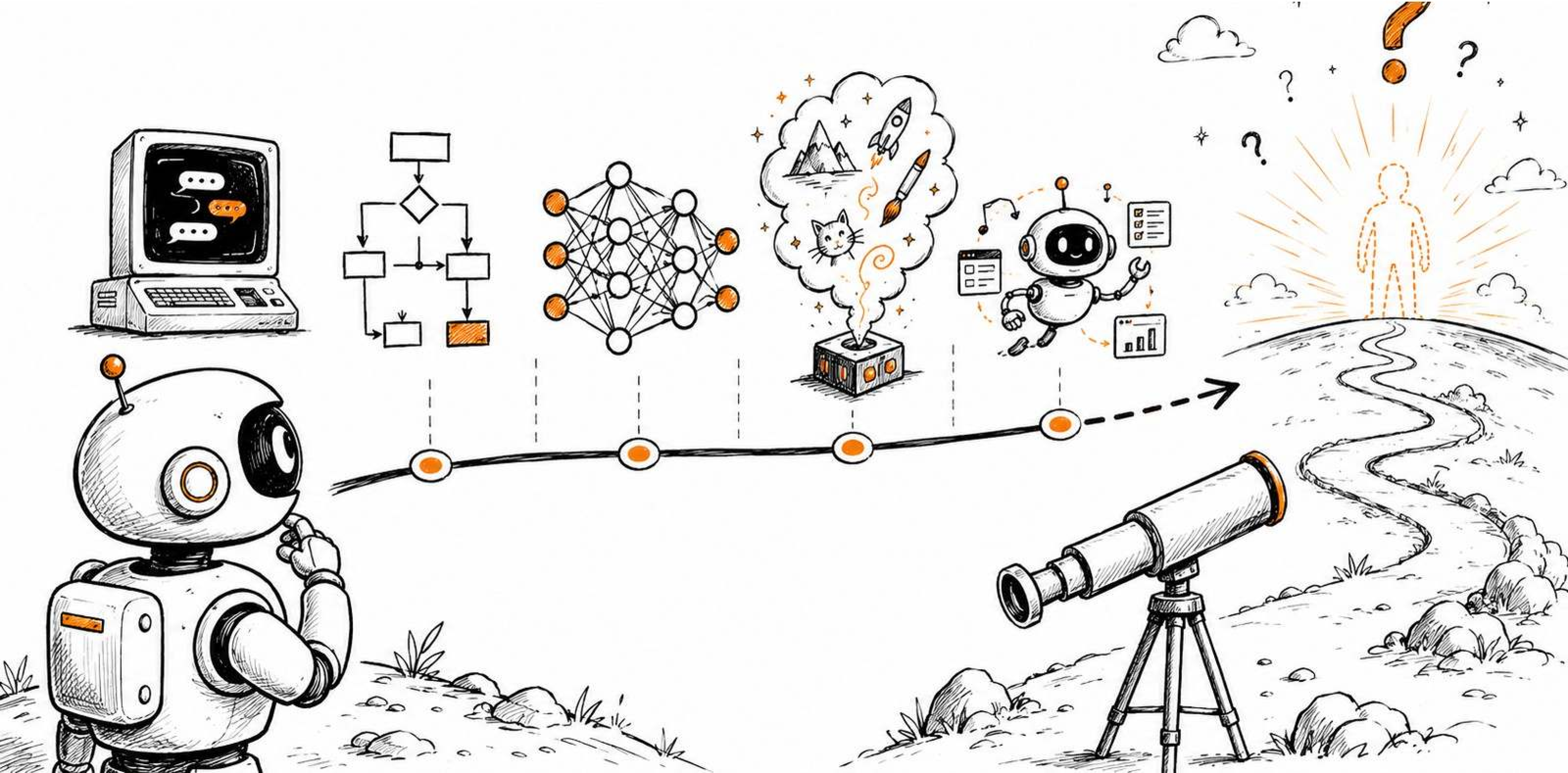
Practical Example: Bank Fraud Detection

- You’re a data scientist at a fintech or bank. Your team is tasked with building an AI model to detec fraudulent credit card transactions in real-time, so that suspicious charges can be blocked or flagged before they cause damage.
- For each AI lifecycle step: (1) Identify a key challenge, (2) Describe the impact on fraud detection, (3 Propose a practical solution

Practical Example: Fraud Detection

Step	Challenge	Business Impact	Proposed Solution
1. Problem Scoping	Misalignment between fraud team and data team on what counts as "fraud"	Wasted time labeling transactions; inconsistent outcomes	Define fraud types clearly with domain experts; align on KPIs (e.g., % fraud blocked)
2. Data Acquisition	Limited access to real fraud examples due to privacy concerns (GDPR)	Incomplete training data reduces model reliability	Use anonymized data; synthetic data generation for rare fraud types
3. Data Exploration	Highly imbalanced data (99.9% legit vs 0.1% fraud)	Model ignores fraud and classifies everything as normal	Use resampling, anomaly detection, or cost-sensitive learning
4. Modeling	Fraud patterns change quickly	Static models become obsolete fast	Use online learning models or retrain frequently
5. Model Evaluation	Using accuracy instead of precision/recall/F1	Model may flag too many false positives or miss real fraud	Focus on recall to catch fraud and precision to avoid false alarms
6. Deployment	High latency from complex models	Delayed fraud alerts; customer dissatisfaction	Optimize for low-latency inference or use lightweight models for real-time use
7. Monitoring & Maintenance	Fraud tactics evolve (new scams & phishing)	Model loses effectiveness over time	Monitor performance, retrain weekly, include human-in-the-loop review system

AI Future: AGI, Agents and What Comes Next



AI History

1966

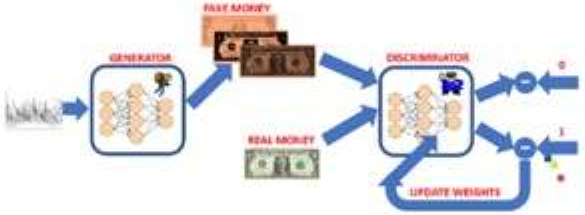
MIT developed the first chatbot Eliza

1989

Yann LeCun Developed Convolutional Neural Networks (CNNs) that can classify images

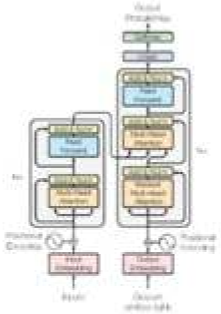
2014

Ian Goodfellow developed Generative Adversarial Networks (GANs) that can generate new images



2017

Transformers were developed by Google



2022

ChatGPT release, AI became mainstream

2023

Microsoft Released CoPilot

2025 & Beyond

Multimodal AI, Agents, Gemini, Sora and video generation AI tools

Future of Artificial General Intelligence (AGI)

- AGI is a type of AI that can outperform humans on most tasks.
- OpenAI predicts to achieve level 5 in 10 years or less.

Level 1: Conversational AI
AI that interacts in conversational language with people. Ex: ChatGPT & Claude

Level 2: Reasoning AI
Performs basic problem-solving tasks comparable to a PhD-level human but without access to any tools.

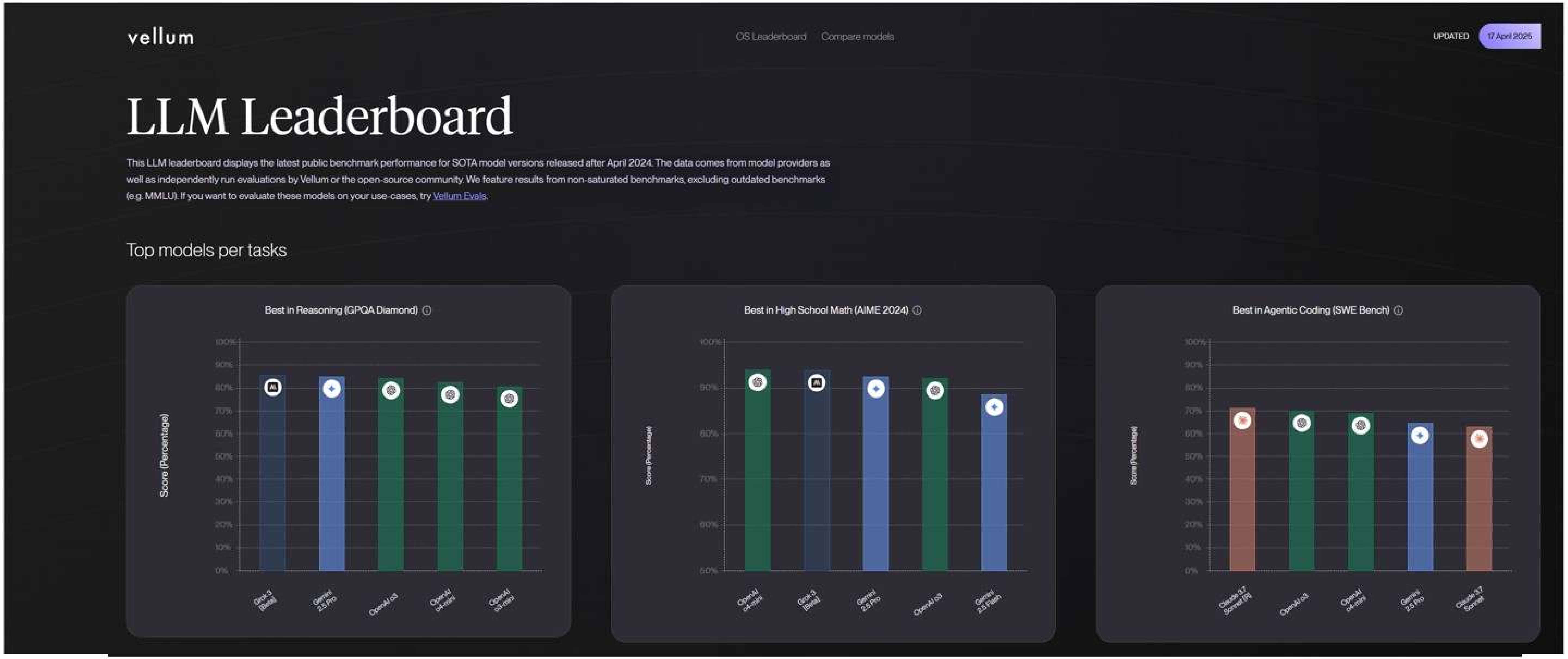
Level 3: Autonomous AI
AI “agents” that can operate autonomously on a user’s behalf.

Level 4: Innovating AI
AI systems that develop innovations independently. Not just running processes but improving them.

Level 5: Organizational AI
Final stage of super AI that performs the work of an organization. Functions are carried out by agents that make improvements without humans.

Is OpenAI Prediction Accurate?

- Visit Vellum: https://www.vellum.ai/llm-leaderboard



Intelligence Beyond AGI

Aspect	ANI (Artificial Narrow Intelligence)	AGI (Artificial General Intelligence)	ASI (Artificial Superintelligence)
Stage	Today	To Come	Post-AGI
Feature Description	Executes specific tasks; cannot generalize to other tasks.	Performs broad tasks, reasons, and improves capabilities comparable to humans.	Demonstrates intelligence beyond human capabilities.
Capabilities	Limited to specific, pre-defined functions (e.g., driving, diagnosing).	Can learn new tasks and reason like a human.	Surpasses human intelligence, making decisions beyond human reasoning.
Examples	Virtual assistants, recommendation systems, autonomous vehicles.	Hypothetical at present; able to pass the Turing Test and perform any intellectual task.	Hypothetical; could innovate independently or tackle complex global issues.
Impact	Outperforms humans in repetitive, specific tasks.	Competes with humans across all intellectual domains.	Superior to humans, capable of solving or threatening societal goals.
Potential Risks	Minimal, as its functions are narrow and controlled.	Medium, depending on its alignment with human values.	High, as its superintelligence could surpass human control.

Summary

AI is the science of creating systems that can mimic human intelligence. This includes general AI, generative AI (which can create new content), & AI agents (autonomous systems that perform complex tasks).

The AI project lifecycle includes 7 phases: problem scoping, data acquisition, data exploration, modeling, evaluation, deployment, & maintenance.

Learners build a simple image classification model using Google Teachable Machine & apply AI lifecycle steps to problems like traffic sign recognition & diabetes prediction.

Common challenges include data quality, model bias, real-time deployment issues, & monitoring.

AI is evolving from narrow AI to AGI and potentially ASI, with predictions that AGI could be achieved within the next decade.